Filter Types and Expressions in Google Earth Engine

Sentinel-2 ImageCollection filtering: date, location, metadata, and filter chaining.

1. Date Filter (Temporal Filter)

Select satellite images acquired within a specified date range.

Expression:

```
ee.Filter.date('2019-01-01', '2020-01-01')
```

Example:

```
var filtered = s2.filter(
  ee.Filter.date('2019-01-01', '2020-01-01')
);
```

2. Location Filter (Spatial Filter)

Select satellite images that intersect the study area (AOI).

Expression:

```
ee.Filter.bounds(geometry)
```

Example:

```
var filtered2 = filtered.filter(
  ee.Filter.bounds(geometry)
);
```

3. Metadata Filter (Cloud Coverage)

Select images with cloud coverage less than or equal to 10%.

Expression:

```
ee.Filter.lte('CLOUDY_PIXEL_PERCENTAGE', 10)
```

Example:

```
var filtered3 = filtered2.filter(
  ee.Filter.lte('CLOUDY_PIXEL_PERCENTAGE', 10)
);
```

4. Filter Chaining

Apply the date, spatial, and cloud coverage filters together.

Example:

```
var filtered = s2
  .filter(ee.Filter.date('2019-01-01', '2020-01-01'))
  .filter(ee.Filter.bounds(geometry))
  .filter(ee.Filter.lte('CLOUDY_PIXEL_PERCENTAGE', 30));

print(filtered);
```

Summary

Filter type	Expression	Function
Date	ee.Filter.date()	Filters images by acquisition date
Location	ee.Filter.bounds()	Filters images by study area
Metadata	ee.Filter.lte()	Filters images by cloud coverage
Chaining	.filter().filter().filter()	Combines all three filters